United States Patent Application Publication 20250287121
Kind Code A1
Publication Date September 11, 2025
Inventor(s) You; Bing Cheng et al.

INTEGRATED CAPACITOR FOR IMAGE SENSOR

Abstract

Some embodiments relate to a device, including: a photodetector; and a pixel circuit, comprising: a floating diffusion node and an output node; a transfer transistor electrically coupled from the floating diffusion node to the photodetector; a source follower transistor comprising a gate electrode electrically coupled to the floating diffusion node; a row select transistor electrically coupled from a source/drain region of the source follower transistor to the output node; and a capacitor electrically coupled from the output node to the floating diffusion node.

Inventors: **You; Bing Cheng (Taichung City, TW), Hung; Feng-Chi (Chu-Bei City, TW), Chen; Hsin-Hung (Tainan City, TW), Hsu; Wen-I (Tainan City, TW), Liu; Jen-Cheng (Hsin-Chu City, TW), Wang; Tzu-Jui (Fengshan City, TW)**

Applicant: **Taiwan Semiconductor Manufacturing Company, Ltd.** (Hsinchu, TW)

Family ID: **1000008064492**

Appl. No.: **18/796403**

Filed: **August 07, 2024**

Related U.S. Application Data

us-provisional-application US 63562860 20240308

Publication Classification

Int. Cl.: **H04N25/771** (20230101); **H01L27/146** (20060101)

U.S. Cl.:

CPC **H04N25/771** (20230101); **H10F39/014** (20250101); **H10F39/8037** (20250101); **H10F39/811** (20250101);

Background/Summary

REFERENCE TO RELATED APPLICATION [0001] This application claims the benefit of U.S. Provisional Application No. 63/562,860, filed on Mar. 8, 2024, the contents of which are incorporated by reference in their entirety.

BACKGROUND
[0002] Integrated circuits (ICs) with image sensors are used in a wide range of modern-day electronic devices, such as, for example, cameras, cellphones, and the like. Types of image sensors include, for example, complementary metal-oxide semiconductor (CMOS) image sensors and charge-coupled device (CCD) image sensors. Compared to CCD image sensors, CMOS image sensors are increasingly favored due to low power consumption, small size, fast data processing, a direct output of data, and low manufacturing cost.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
[0004] FIGS. **1**A, **1**B, and **1**C illustrate circuit schematics of some embodiments of an image sensor with a capacitor between an output node and a floating diffusion node.
[0005] FIG. **2** illustrates a cross-sectional view of an output stage of some embodiments of an image sensor with a capacitor between the output node and the floating diffusion node.
[0006] FIG. **3** illustrates a circuit schematic of some embodiments of an image sensor with a capacitor between the output node and the floating diffusion node with multiple photodetectors coupled to the floating diffusion node.
[0007] FIG. **4** illustrates a cross-sectional view of some embodiments of an image sensor with a shield structure and a metal bond pad configured to be a capacitor coupled between the output node and the floating diffusion node.
[0008] FIGS. **5**A and **5**B illustrate top down views of some embodiments of a plurality of shield structures and metal bond pads in an array of pixel circuits.
[0009] FIGS. **6-18** illustrate a series of cross-sectional views of some embodiments of a method of forming an image sensor with a capacitor between the output node and the floating diffusion node with multiple photodetectors coupled to the floating diffusion node.
[0010] FIG. **19** illustrates a flowchart of some embodiments of a method of forming an image sensor with a capacitor between the output node and the floating diffusion node.
DETAILED DESCRIPTION
[0011] The present disclosure provides many different embodiments, or examples, for implementing different features of this disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments

and/or configurations discussed.

[0012] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0013] It will be appreciated that in this written description, as well as in the claims below, the terms “first”, “second”, “second”, “third” etc. are merely generic identifiers used for ease of description to distinguish between different elements of a figure or a series of figures. In and of themselves, these terms do not imply any temporal ordering or structural proximity for these elements, and are not intended to be descriptive of corresponding elements in different illustrated embodiments and/or un-illustrated embodiments. For example, “a first dielectric layer” described in connection with a first figure may not necessarily correspond to a “first dielectric layer” described in connection with another figure, and may not necessarily correspond to a “first dielectric layer” in an un-illustrated embodiment.

[0014] An image sensor comprises a pixel array with a plurality of photodetectors and a plurality of pixel circuits coupled to the photodetectors. The plurality of pixel circuits comprise a floating diffusion node, a transfer transistor extending between the floating diffusion node and the photodetector, a reset transistor with a source/drain terminal coupled to the floating diffusion node, and an output stage coupled to the floating diffusion node. The output stage comprises a source follower transistor with a gate electrode coupled to the floating diffusion node and further comprises a row select transistor with a first source/drain terminal coupled to a source/drain terminal of the source follower transistor. An output node is at a second source/drain terminal of the row select transistor. The plurality of pixel circuits is organized in a plurality of rows and columns, and each column of the plurality of pixel circuits have output nodes coupled together by an output line.

[0015] In some embodiments, the image sensor spans multiple substrates bonded together through bond layers. The reset transistor and the output stage are on a first substrate. The photodetector and the floating diffusion node are within a second substrate coupled to the first substrate by a first bond layer. The floating diffusion node is coupled to the output stage through a metal bond pad within the first bond layer. A shield structure is also within the first bond layer and continuously surrounds the metal bond pad.

[0016] A total parasitic capacitance C.sub.tot at the output line is a sum of pixel parasitic capacitance values C.sub.js between output nodes of the plurality of pixel circuits coupled to an output line. The total capacitance C.sub.tot increases an RC delay of the image sensor. The increased RC delay results in a greater amount of time taken to transfer the signals from the pixel circuit to an image signal processor (ISP) circuit. The gain of the image sensor in low information environments (e.g., when using a conversion gain circuit to image areas with low amounts of light) are also reduced due to the total capacitance C.sub.tot. The RC delay further results in a reduction in bandwidth of the image sensor. Therefore, a method of mitigating the total capacitance C.sub.tot is desirable.

[0017] The present disclosure provides for a capacitor being included between the output node and the floating diffusion node. Applying the Miller theorem across the capacitor, the output stage having a gain of less than one results in the capacitance C.sub.H of the capacitor as viewed from the output node appearing to be negative. Therefore, the combination of the pixel parasitic capacitance value C.sub.js and the C.sub.H value for a pixel circuit is less than the pixel parasitic capacitance value C.sub.js alone. Applying this result across the pixel circuits coupled to the output line results in a reduction in the perceived capacitance as viewed from the output of the circuit.

This reduction in perceived capacitance reduces the RC delay of the image sensor, increasing the gain in low information applications (e.g., imaging areas with low amounts of light) and the bandwidth of the image sensor. The increase in the gain in low information applications results in an increase in the dynamic range of the image sensor and consequently an increase in the quality of images produced by the image sensor.

[0018] FIGS. **1**A, **1**B, and **1**C illustrate circuit schematics **100***a*, **100***b*, **100***c* of some embodiments of an image sensor with a capacitor between an output node and the floating diffusion node.

[0019] A photodetector **102** is coupled to a pixel circuit **104**. The pixel circuit **104** comprises a floating diffusion node **106**, a transfer transistor **108** extending between the floating diffusion node **106** and the photodetector **102**, a reset transistor **110** with a source/drain terminal **112** coupled to the floating diffusion node **106**, and an output stage **114** coupled to the floating diffusion node **106**. The output stage **114** comprises a source follower transistor **116** with a first gate electrode **118** coupled to the floating diffusion node **106** and a row select transistor **120** with a first source/drain terminal **122** coupled to a second source/drain terminal **124** of the source follower transistor **116**. A third source/drain terminal **126** of the source follower transistor is coupled to a voltage supply rail Vdd. An output node **128** is coupled to a fourth source/drain terminal **130** of the row select transistor **120**. An output line **140** is coupled to the output node **128**, and a row select line **142** is coupled to a gate electrode of the row select transistor **120**.

[0020] The output node **128** has a parasitic capacitance C.sub.js that is represented by a first capacitor **132**. A second capacitor **134** is coupled between the output node **128** and the floating diffusion node **106** and has a capacitance C.sub.H. The capacitance C.sub.H and the parasitic capacitance C.sub.js both affect the perceived capacitance at the output node **128**. The perceived capacitance due to the second capacitor **134** depends on a ratio of a voltage at the output node **128** to a voltage at the floating diffusion node **106**. The ratio of the voltage at the output node to the voltage at the floating diffusion node **106** may also be represented as the gain of the output stage **114**.

[0021] When the gain of the output stage **114** is greater than one (e.g., when the voltage at the output node **128** is greater than the voltage at the floating diffusion node **106**), the perceived capacitance of the second capacitor **134** is positive, and the total perceived capacitance at the output node is greater than the capacitance at the output node supplied by the parasitic capacitance C.sub.js. When the gain of the output stage **114** is less than one (e.g., when the voltage at the output node **128** is less than the voltage at the floating diffusion node **106**), the perceived capacitance of the second capacitor **134** is negative, and the total perceived capacitance at the output node is less than the capacitance at the output node **128** supplied by the parasitic capacitance C.sub.js (see FIG. **1**B for greater detail). The reduction in the total capacitance C.sub.tot perceived at the output node **128** reduces the RC delay in the image sensor, which increases the gain in low information applications. The gain is increased due to the reduction in time constant of the pixel array, resulting in increasing the range of magnitudes of the signal available within a measurement timeframe and the bandwidth of the image sensor (e.g., by reducing the time constant, reducing the minimum measurement timeframe to transfer the signals from the pixel circuits to the ISP circuit). The increase in the gain in low information applications results in an increase in the dynamic range of the image sensor and an increase in the quality of images produced by the image sensor.

[0022] As shown in the circuit schematic **100***b* of FIG. **1**B, an equivalent circuit to the one shown in the circuit schematic **100***a* of FIG. **1**A is provided. The second capacitor (see **134** of FIG. **1**A) may be treated as a third capacitor **136** coupled between the floating diffusion node **106** and ground, and a fourth capacitor **138** coupled between the output node **128** and ground. This results in a circuit where the perceived capacitance at the output node **128** can be determined by adding the capacitance of the first capacitor **132** to the capacitance of the fourth capacitor **138**, for each pixel circuit **104** of the plurality of pixel circuits coupled to the output line **140**. The capacitance C.sub.4 of the fourth capacitor **138** may be determined based on the capacitance C.sub.H of the second capacitor (see **134** of FIG. **1**A) by using Miller's theorem (substituting the initial impedance Z of Miller's Theorem with the impedance of the second capacitor and the output impedance Z.sub.2 of Miller's Theorem with the impedance of the fourth capacitor):

[00001] $\frac{1}{jwC_4} = \frac{\frac{1}{jwC_H}}{1 - \frac{1}{k}}$ (1)

Where k is the ratio of the voltage at the output node **128** to the voltage at the floating diffusion node **106** (e.g., the gain of the output stage **114**). Equation (1) may be simplified and solved for C.sub.4, resulting in Equation (2) shown below:

[00002] $C_4 = C_H(1\frac{1}{k})$ (2)

[0023] As shown in Equation (2), when k is greater than one, the factor that the capacitance C.sub.H is multiplied by to derive C.sub.4 is positive, leading to C.sub.4 being positive. Further, when k is less than one, the factor that the capacitance C.sub.H is multiplied by to derive C.sub.4 is negative, resulting in C.sub.4 being negative. As the first capacitor **132** (representing the parasitic capacitance C.sub.js) and the fourth capacitor **138** are in parallel in the equivalent circuit shown in FIG. **1**B, the total perceived capacitance at the output node **128** may be found by finding the sum of all the capacitances coupled to the output node **128**. In the pixel circuit shown in FIG. **1**B, when k is less than one, the sum of the parasitic capacitance C.sub.js and the capacitance C.sub.4 is less than the parasitic capacitance C.sub.js, resulting in a reduction in the perceived capacitance at the output node **128**.

[0024] As shown in the circuit schematic **100***c* of FIG. **1**C, a plurality of pixel circuits **145** is organized into a plurality of rows **146**, **148** and columns **150**, **152**. A first row select line **142***a* is coupled to gate electrodes **144** of row select transistors **120** in a first row **146** of pixel circuits. A second row select line **142***b* is coupled to gate electrodes **144** of row select transistors **120** in a second row **148** of pixel circuits. A first output line **140***a* is coupled to output nodes **128** of a first column **150** of pixel circuits. A second output line **140***b* is coupled to output nodes **128** of a second column **152** of pixel circuits.

[0025] The perceived capacitance at the first output line **140***a* is affected by the parasitic capacitance C.sub.js and the capacitance C.sub.H of each pixel circuit in the first column **150**. The total capacitance C.sub.tot perceived at the output line can be found by finding the sum of the parasitic capacitance and the capacitance C.sub.4 where an equivalent circuit as shown in FIG. **1**B is used in place of the pixel circuits **104** of FIG. **1**C. The total capacitance C.sub.tot is found using Equation 3 shown below:

[00003] $C_{tot} = n \star C_{js} + n \star (C_H(1\frac{1}{k}))$ (3)

Where n is the total number of pixel circuits in the first column **150**. For example, when the gain of the output stage of the pixel circuits is 0.9, one ninth of the capacitance C.sub.H is subtracted from the total capacitance C.sub.tot perceived at the first output line **140***a*. The capacitance C.sub.H and the gain across the capacitors per pixel circuit **104** in the first column **150** also are included in C.sub.tot.

[0026] FIG. **2** illustrates a cross-sectional view **200** of an output stage of some embodiments of an image sensor with a capacitor between an output node and the floating diffusion node.

[0027] The parasitic capacitance C.sub.js (as represented by the first capacitor **132**) is between the fourth source/drain terminal **130** of the row select transistor and a body region of a first substrate **202**. A first electrode **204** of the second capacitor **134** is coupled to the fourth source/drain terminal **130** by a first contact **206**. An interconnect structure **208** further extends between and electrically couples the first electrode **204** and the first contact **206**, resulting in coupling the first electrode **204** to the output node **128**. A second electrode **210** of the second capacitor **134** is coupled to the first gate electrode **118** of the source follower transistor **116** by the interconnect structure **208** and a second contact **212**, where the second electrode **210** is coupled to the floating diffusion node (see **106** of FIG. **1**A). The interconnect structure **208** comprises one or more wire levels and one or more via levels forming a first conductive path **214** between the first gate electrode **118** and the second electrode **210** and a second conductive path **216** between the fourth source/drain terminal **130** and the first electrode **204**. The interconnect structure **208** is surrounded by a plurality of interlayer dielectric layers **218**. A first insulative layer **220** surrounds the first and second electrodes **204**, **210** of the second capacitor **134**.

[0028] FIG. **3** illustrates a circuit schematic **300** of some embodiments of an image sensor with a capacitor between an output node and the floating diffusion node with multiple photodetectors coupled to the floating diffusion node.

[0029] In some embodiments, the pixel circuit **104** is formed across multiple different chips bonded together. In some embodiments, the reset transistor **110**, the source follower transistor **116**, and the row select transistor **120** are on a first chip **306**. A plurality of photodetectors **302** are coupled to the floating diffusion node **106** by a plurality of transfer transistors **304** on a second chip **308**. An ISP circuit **310** is on a third chip **312**. The row select line **142** and the output line **140** are within the first chip **306**. The output node **128** is on the first chip **306** and is electrically coupled to the third chip **312**. In some embodiments, the first electrode **204** and the second electrode **210** of the second capacitor **134** are within bond layers between the second chip **308** and the first chip **306**.

[0030] FIG. **4** illustrates a cross-sectional view **400** of some embodiments of an image sensor with a shield structure and a metal bond pad configured to be a capacitor coupled between an output node and the floating diffusion node.

[0031] In some embodiments, the first interconnect structure **208** is on a first side **202***a* of the first substrate **202** and comprises the first conductive path **214**. The first conductive path **214** electrically couples the first gate electrode **118** of the source follower transistor **116** to the second electrode **210** of the second capacitor **134**. The first conductive path **214** further is electrically coupled to the source/drain terminal **112** of the reset transistor **110** and a conversion gain circuit **402**. The first conductive path **214** is coupled to the floating diffusion node **106**. The conversion gain circuit **402** comprises one or more semiconductor devices configured to increase the conversion gain of the pixel circuit in a low information environment (e.g., imaging an area with a light level below a specified threshold).

[0032] The first conductive path **214** is coupled to a third conductive path **404** in a second interconnect structure **406** on the second chip **308**. A first bond layer **408** of the first chip **306** and a second bond layer **411** of the second chip **308** mechanically couple the first chip **306** to the second chip **308**. The second electrode **210** comprises the combination of a first metal bond pad **405***a* of a first plurality of metal bond pads **405** within the first bond layer **408** and a second metal bond pad **415***a* of a second plurality of metal bond pads **415** within the second bond layer **411**. The second metal bond pad **415***a* is coupled to the first metal bond pad **405***a* at a first bond interface **421**. The second electrode **210** electrically couples the first conductive path **214** to the third conductive path **404**. The second electrode **210** is also referred to as the bond electrode. Further, a first insulative layer **220** of the first bond layer **408** is mechanically coupled to a second insulative layer **423** of the second bond layer **411**. The second interconnect structure **406** further comprises one or more additional conductive paths **416** coupled to the plurality of transfer transistors **304**. Separate conductive paths **416** for the plurality of transfer transistors **304** coupled to the floating diffusion node **106** results in the transfer of charge from individual photodetectors of the plurality of photodetectors **302** during operation so that the charge from individual photodetectors can be separately delivered to the pixel circuit **104**.

[0033] The second conductive path **216** extends from the fourth source/drain terminal **130** of the row select transistor **120** to the first electrode **204**. In some embodiments, the second conductive path **216** is the same as the output node **128**. In some embodiments, the first electrode **204** surrounds the second electrode **210**, increasing the surface area of the second capacitor **134** and insulating the second electrode **210** from electric fields caused by other metal bond pads of the first plurality of metal bond pads **405**. The first electrode **204** comprises a combination of a first shield structure **407***a* of a first plurality of shield structures **407** and a second shield structure **413***a* of a second plurality of shield structures **413**. The first electrode **204** is also referred to as the shield electrode. The first shield structure **407***a* is mechanically coupled to the second shield structure **413***a* of a second plurality of shield structures **413** at the first bond interface **421**. The second shield structure **413***a* has a same layout as the first shield structure **407***a* (e.g., has a same length and width, and a same distance between inner sidewalls) to enhance the bond strength at the first bond interface **421**. The combination of the first bond layer **408** and the second bond layer results in an increased surface area of inner sidewalls of the first electrode **204** and outer sidewalls of the second electrode **210**. The increased surface area increases the capacitance of the second capacitor **134** without increasing the area used by the second capacitor **134** within the first or second interconnect structures **208**, **406**. That is, a thickness of the first electrode **204** and the second electrode **210** is greater than a second thickness of the wire levels of the interconnect structure, resulting in a greater capacitance between the first electrode and the second electrode than a capacitance that would be between wires in a same layout to that of the first electrode **204** and the second electrode **210**.

[0034] The second conductive path **216** is also electrically coupled to the ISP circuit **310** on the third chip **312**. The ISP circuit **310** is configured to process the signals passed to it from the plurality of pixel circuits (see **145** of FIG. **1**C) in the image sensor in order to combine the signals into an image. The ISP circuit **310** comprises one or more correlated double sampling (CDS) circuits, analog to digital converter (ADC) circuits, and amplifier circuits. One or more through substrate vias (TSVs) **410** couple the first interconnect structure **208** to a third interconnect structure **412** on a second side **202***b* of the first substrate **202** opposite the first side **202***a*. In some embodiments, third interconnect structure **412** comprises one or more wire levels and one or more via levels that are electrically coupled to the third chip **312** by a third bond layer **409**. The third bond layer **409** comprises a third plurality of metal bond pads **414** and a third insulative layer **425**. The third bond layer **409** is electrically coupled to a fourth bond layer **417** of the third chip **312** at a second bond interface **426**. A fourth plurality of metal bond pads **419** are mechanically and electrically coupled to the third plurality of metal bond pads **414**. Further, a fourth insulative layer **427** is mechanically coupled to the third insulative layer **425**. The third plurality of metal bond pads **414** and the fourth plurality of metal bond pads **419** electrically couple the third interconnect structure **412** to the ISP circuit **310**.

[0035] The plurality of photodetectors **302** are positioned within a second substrate **418**. A plurality of deep trench isolation (DTI) structures **420** extend around and isolate the plurality of photodetectors from each other. A plurality of color filters **422** and micro lenses **424** are positioned above the photodetectors **302**, such that incoming light is filtered and directed towards the photodetectors **302** before entering the second substrate **418**.

[0036] FIGS. **5**A and **5**B illustrate top down views **500***a*, **500***b* of some embodiments of a plurality of shield structures and metal bond pads in an array of pixel circuits.

[0037] As shown in the top down view **500***a* of FIG. **5**A, the first bond layer **408** comprises the first insulative layer **220** surrounding a first plurality of metal bond pads **405** including the first metal bond pad **405***a* and a first plurality of shield structures **407** including the first shield structure **407***a*. In some embodiments, the first plurality of metal bond pads **405** are respectively surrounded by the first plurality of shield structures **407**. For example, in an image sensor with four pixel circuits, the first metal bond pad **405***a* is surrounded by the first shield structure **407***a*, a second metal bond pad **405***b* is continuously surrounded by a second shield structure **407***b*, a third metal bond pad **405***c* is continuously surrounded by a third shield structure **407***c*, and a fourth metal bond pad **405***d* is continuously surrounded by a fourth shield structure **407***d*. The number of metal bond pads in the first plurality of metal bond pads **405** is equal to the number of shield structures in the first plurality of shield structures **407**.

[0038] As shown in the top down view **500***b* of FIG. **5**B, in other embodiments, the number of metal bond pads in the first plurality of metal bond pads **405** is greater than the number of shield structures in the first plurality of shield structures **407**. The shield structures surround more than one metal bond pad within columns of the plurality of metal bond pads. For example, in an image sensor with four pixel circuits, the first metal bond pad **405***a* and the second metal bond pad **405***b* in the first column **150** are continuously surrounded by the first shield structure **407***a*, and the third metal bond pad **405***c* and fourth metal bond pad **405***d* in the second column **152** are surrounded by the second shield structure **407***b*. The output nodes (see **128** of FIG. **1**C) of the plurality of pixel circuits in the first column **150** are coupled together by the output line (see **140** of FIG. **1**C). As the output nodes (see **128** of FIG. **1**C) are also coupled to the shield structures of the first plurality of shield structures **407** in the first column **150**, the electrically coupled shield structures shown in FIG. **1**A can be replaced by the shield structures shown in FIG. **1**B without changing the electrical couplings of the circuit. Using the shield structures as shown in FIG. **5**B reduces the footprint of the shield structures (e.g., the lateral area of the shield structure may be reduced). By removing portions of the shield structure from between the metal bond pads, the metal bond pads may be positioned closer together, reducing the footprint of the pixel circuits.

[0039] FIGS. **6-18** illustrate a series of cross-sectional views **600-1800** of some embodiments of a method of forming an image sensor with a capacitor between an output node and the floating diffusion node with multiple photodetectors coupled to the floating diffusion node. Although FIGS. **6-18** are described as a series of acts, it will be appreciated that these acts are not limiting in that the order of the acts can be altered in other embodiments, and the methods disclosed are also applicable to other structures. In other embodiments, some acts that are illustrated and/or described may be omitted in whole or in part.

[0040] As shown in the cross-sectional view **600** of FIG. **6**, the third interconnect structure **412** is formed on the second side **202***b* of the first substrate **202** as part of the first chip **306**. The third interconnect structure **412** comprises one or more wire levels and one or more via levels forming conductive paths over the first substrate **202**. Further, a plurality of interlayer dielectric layers **218** are formed between forming wire levels and via levels of the third interconnect structure **412**. In some embodiments, the third interconnect structure **412** comprises a conductive material, such as copper, aluminum, tungsten, a conductive metal alloy, or the like. In some embodiments, the plurality of interlayer dielectric layers **218** are or comprise an insulative material, such as silicon dioxide (SiO.sub.2), silicon nitride (Si.sub.3N.sub.4), or the like. The third interconnect structure **412** is formed using one or more of a physical vapor deposition (PVD), atomic layer deposition (ALD), chemical vapor deposition (CVD), a damascene process, a dual damascene process, or the like.

[0041] As shown in the cross-sectional view **700** of FIG. **7**, in some embodiments, a third bond layer **409** is formed on the third interconnect structure **412**. The third bond layer **409** comprises a third insulative layer **425** and a third plurality of metal bond pads **414**. In some embodiments, the third plurality of metal bond pads **414** are or comprise one or more of aluminum, copper, aluminum copper, or the like. In some embodiments, the third insulative layer **425** is or comprises one or more of silicon dioxide (SiO.sub.2), silicon nitride (Si.sub.3N.sub.4), or the like. The third insulative layer **425** is formed using one or more of PVD, ALD, CVD, or the like. The third plurality of metal bond pads **414** are formed using one or more of PVD, ALD, CVD, a damascene process, or the like. In some embodiments, the third plurality of metal bond pads **414** are formed concurrently with an underlying contact layer by using a dual damascene process.

[0042] As shown in the cross-sectional view **800** of FIG. **8**, the ISP circuit **310** is formed on the third chip **312** and the third chip **312** is bonded to the first chip **306** at the third bond layer **409**. The ISP circuit **310** comprises one or more correlated double sampling (CDS) circuits, analog to digital converter (ADC) circuits, and amplifier circuits which comprise one or more transistors, passive circuit components, or other circuit components. Forming the ISP circuit **310** comprises using one or more of PVD, ALD, CVD, and implantation processes or the like to form the transistors and circuit components, as well as using one or more of PVD, ALD, CVD, damascene processes, or the like to form an overlying interconnect structure to form conductive paths between the circuit components and the transistors.

[0043] In some embodiments, the third chip **312** is coupled to the first chip **306** by forming the fourth bond layer **417** on the third chip **312** before bonding the third chip **312** to the third bond layer **409**. The fourth insulative layer **427** is dielectrically bonded to the third insulative layer **425**, and the fourth plurality of metal bond pads **419** are bonded to the third plurality of metal bond pads **414**. The combination of using a dielectric-to-dielectric bond between insulative layers and a metal-to-metal bond between metal bond pads demonstrates a hybrid bond at the second bond interface **426**. The dielectric-to-dielectric bond between insulative layers and a metal-to-metal bond between metal bond pads at the second bond interface **426** is formed by performing a pressure treatment to initially bond the fourth insulative layer **427** to the third insulative layer **425** and a subsequent anneal to strengthen the bond between the third and fourth insulative layers **425**, **427**, as well as forming the bond between the third and fourth pluralities of metal bond pads **414**, **419**. In other embodiments, a hybrid bond is not used to bond the first chip **306** to the third chip **312**, and a different method of bonding the first chip **306** to the third chip **312** is performed.

[0044] As shown in the cross-sectional view **900** of FIG. **9**, a plurality of front end of line (FEOL) circuit components **902** are formed on the first side **202***a* of the first substrate **202**. The plurality of FEOL circuit components **902** comprise reset transistors **110**, components of the conversion gain circuit **402**, source follower transistors **116**, and row select transistors **120** of the plurality of pixel circuits. In some embodiments the plurality of FEOL circuit components **902** formed within the image sensor are a plurality of metal oxide semiconductor field effect transistor (MOSFET) devices. In some embodiments, the plurality of FEOL circuit components **902** are formed using one or more PVD, ALD, CVD, or implantation processes.

[0045] As shown in the cross-sectional view **1000** of FIG. **10**, contacts **1002**, including the first contact **206** and the second contact **212**, and an overlying wire layer **1004** are formed over the first side of the first substrate **202**. The first contact **206** is coupled to the fourth source/drain terminal **130** of the row select transistor **120**, and the second contact **212** is coupled to a first gate electrode **118** of the source follower transistor **116**. The contacts **1002** are coupled to the plurality of FEOL circuit components **902** to connect them to the first interconnect structure (see **208** of FIG. **11**) to be formed hereafter. Further, the TSV **410** is formed, coupling the fourth source/drain terminal **130** of the row select transistor **120** (via the overlying wire layer **1004**) to the third interconnect structure **412** and the ISP circuit **310**. In some embodiments, the contacts **1002**, the TSV, and the overlying wire layer **1004** are formed using one or more of etching, CVD, ALD, PVD, and planarization (e.g., chemical mechanical planarization) processes. In some embodiments, the contacts **1002** and the overlying wire layer **1004** are formed concurrently.

[0046] As shown in the cross-sectional view **1100** of FIG. **11**, the first interconnect structure **208** is formed over the contacts **1002**. The first interconnect structure **208** comprises a plurality of wire levels and a plurality of via levels arranged to form at least the first conductive path **214** and the second conductive path **216**. The first conductive path **214** electrically couples the first gate electrode **118** of the source follower transistor **116**, a source/drain terminal of the reset transistor **110**, and a transistor of the conversion gain circuit **402**, and extends to an uppermost wire level of the first interconnect structure. The second conductive path **216** is coupled to the fourth source/drain terminal **130** of the row select transistor **120** and the ISP circuit **310**, and extends to an uppermost wire level of the first interconnect structure **208**. In some embodiments, the overlying wire layer (see **1004** of FIG. **10**) is part of the first interconnect structure **208**.

[0047] As shown in the cross-sectional view **1200** of FIG. **12**, the first bond layer **408** is formed over and coupled to the first interconnect structure **208**. The first bond layer **408** comprises a first insulative layer **220**, a first plurality of metal bond pads **405** (including the second electrode **210**) and a first plurality of shield structures **407** (including the first electrode **204**), as shown in the top layouts of FIGS. **5**A and **5**B. The spacing of the first plurality of metal bond pads **405** from the first plurality of shield structures **407** form a plurality of capacitors including the second capacitor **134**. The capacitance of the plurality of capacitors is based on both a distance between opposing faces of the first plurality of metal bond pads **405** (e.g., outer sidewalls of the first plurality of metal bond pads **405**) and the first plurality of shield structures **407** (e.g., inner sidewalls of the first plurality of shield structures **407**) and a surface area of the opposing faces of the first plurality of metal bond pads **405** and the first plurality of shield structures **407**. The surface area of the opposing faces is increased when the second bond layer (see **411** of FIG. **4**) is bonded to the first bond layer **408** (see FIG. **16**). The first plurality of shield structures **407** are coupled to the output nodes **128** of the pixel circuits (see **104** of FIG. **1**A) by the second conductive paths **216** in the first interconnect structure **208**. The first plurality of metal bond pads **405** are coupled to the source follower transistors **116** by the first conductive paths **214**.

[0048] The first plurality of metal bond pads **405** are or comprise one or more of aluminum, copper, aluminum copper, or the like. The first insulative layer **220** is or comprises one or more of silicon dioxide (SiO.sub.2), silicon nitride (Si.sub.3N.sub.4), or the like. The first insulative layer **220** is formed using one or more of PVD, ALD, CVD, etching, or the like. The first plurality of metal bond pads **405** are formed using one or more of PVD, ALD, CVD, a damascene process, or the like. In some embodiments, the first plurality of metal bond pads **405** are formed concurrently with an underlying contact layer by using a dual damascene process.

[0049] As shown in the cross-sectional view **1300** of FIG. **13**, the plurality of transfer transistors **304**, the plurality of photodetectors **302**, and the floating diffusion node **106** are formed on the second substrate **418** of the second chip **308**. The floating diffusion node **106** and the plurality of photodetectors **302** are formed using implantation processes to implant n-type dopants into the second substrate **418**. In some embodiments, the plurality of transfer transistors **304** are formed using one or more of PVD, ALD, CVD, and etching processes.

[0050] As shown in the cross-sectional view **1400** of FIG. **14**, the second interconnect structure **406** is formed over the second substrate **418**. The second interconnect structure **406** forms a third conductive path **404** coupled to the floating diffusion node **106** and additional conductive paths coupled to the gate electrodes of the transfer transistors **304**. The second interconnect structure **406** comprises a plurality of wire levels and a plurality of via levels. A plurality of interlayer dielectric layers **218** surround the second interconnect structure **406** and are formed between the formation of individual wire levels. In some embodiments, the second interconnect structure **406** comprises a conductive material, such as copper, aluminum, aluminum copper, tungsten, a conductive metal alloy, or the like. In some embodiments, the plurality of interlayer dielectric layers **218** are or comprise an insulative material, such as silicon dioxide (SiO.sub.2), silicon nitride (Si.sub.3N.sub.4), or the like. The second interconnect structure **406** is formed using one or more of a PVD, ALD, CVD, damascene processes, dual damascene processes, or the like.

[0051] As shown in the cross-sectional view **1500** of FIG. **15**, in some embodiments, a second bond layer **411** is formed over the second interconnect structure **406** on the second chip **308**. The second bond layer **411** comprises a second insulative layer **423** of a same material as the first insulative layer **220**, a second plurality of shield structures **413** comprising a same material as the first plurality of shield structures **407**, and a second plurality of metal bond pads **415** that comprise a same material as the first plurality of metal bond pads **405**. The second plurality of metal bond pads **415** are arranged to meet and couple to exposed faces of the first plurality of metal bond pads **405** during a bonding process described hereafter (see FIG. **16**). The second plurality of shield structures **413** are arranged to meet and couple to exposed faces of the first plurality of shield structures **407** during the bonding process described hereafter (see FIG. **16**). In some embodiments, the second plurality of metal bond pads **415** has a same top geometry as the first plurality of metal bond pads (see **405** of FIG. **4**), and wherein the second plurality of shield structures **413** has a same top geometry as the first plurality of shield structures (see **407** of FIG. **4**). In other embodiments, the second bond layer **411** is omitted.

[0052] As shown in the cross-sectional view **1600** of FIG. **16**, the plurality of DTI structures **420** are formed within the second substrate **418** around the plurality of photodetectors **302**. In some embodiments, the plurality of DTI structures **420** are formed using one or more of PVD, ALD, CVD, or etching processes. In some embodiments, the plurality of DTI structures **420** are formed after the second chip **308** is bonded to the first chip **306** (e.g., subsequent to the steps shown in FIG. **17**).
[0053] As shown in the cross-sectional view **1700** of FIG. **17**, the second chip **308** is bonded to the first chip **306**, such that the third conductive path **404** is electrically coupled to the first conductive path **214** by the first plurality of metal bond pads **405** (including the second electrode **210**) and the second plurality of metal bond pads **415**. The second chip **308** is bonded to the first chip **306** using a combination of a dielectric-to-dielectric bond between insulative layers and a metal-to-metal bond between metal bond pads. The second insulative layer **423** and the first insulative layer **220** are bonded together using a pressure treatment and a subsequent anneal. Further, the first plurality of metal bond pads **405** bond to the second plurality of metal bond pads **415** and the first plurality of shield structures **407** bind to the second plurality of shield structures **413** during the pressure treatment and subsequent anneal. Bonding the first bond layer **408** to the second bond layer **411** results in the second capacitor **134** having an increased area, increasing the capacitance of the second capacitor **134**.
[0054] As shown in the cross-sectional view **1800** of FIG. **18**, in some embodiments, the plurality of color filters **422** and the plurality of micro lenses **424** are formed on the second substrate **418** over the photodetectors **302**. In some embodiments, the plurality of color filters **422** and the plurality of micro lenses **424** are formed before second chip **308** is bonded to the first chip **306**, subsequent to the forming of the plurality of photodetectors **302** and the plurality of DTI structures **420**. In some embodiments, the plurality of color filters **422** and the plurality of micro lenses **424** are centered on individual photodetectors of the plurality of photodetectors **302**, such that the plurality of micro lenses **424** direct light towards the photodetectors **302**. In other embodiments, the plurality of micro lenses **424** are offset from the center of individual photodetectors based on the position of the individual photodetectors in the photodetector array. For example, in some embodiments, individual photodetectors near a center of the photodetector array will have microlenses closer to being centered on the individual photodetectors, while individual photodetectors near an outer edge of the array will have microlenses with a greater offset from being centered on the individual photodetectors, to more effectively bend the incoming light towards the individual photodetectors.
[0055] FIG. **19** illustrates a flowchart **1900** of some embodiments of a method of forming an image sensor with a capacitor between an output node and a floating diffusion node. Although this method and other methods illustrated and/or described herein are illustrated as a series of acts or events, it will be appreciated that the present disclosure is not limited to the illustrated ordering or acts. Thus, in some embodiments, the acts may be carried out in different orders than illustrated, and/or may be carried out concurrently. Further, in some embodiments, the illustrated acts or events may be subdivided into multiple acts or events, which may be carried out at separate times or concurrently with other acts or sub-acts. In some embodiments, some illustrated acts or events may be omitted, and other un-illustrated acts or events may be included.
[0056] At **1902**, an output stage is formed on a first substrate and comprising a row select transistor electrically coupled from a source follower transistor to an output node. An example of a drawing illustrating this step can be found, for example, in FIG. **9**.
[0057] At **1904**, a first interconnect structure is formed over the output stage, wherein the first interconnect structure comprises a first conductive path electrically coupled to a gate electrode of the source follower transistor and a second conductive path electrically coupled to the output node. An example of a drawing illustrating this step can be found, for example, in FIGS. **10-11**.
[0058] At **1906**, a first bond layer is formed over the first interconnect structure, wherein the first

bond layer comprises a bond electrode and a shield electrode surrounding the bond electrode, and wherein the shield electrode is formed electrically coupled to the second conductive path. An example of a drawing illustrating this step can be found, for example, in FIG. **12**.

[0059] At **1908**, a photodetector is formed in a second substrate. An example of a drawing illustrating this step can be found, for example, in FIG. **13**.

[0060] At **1910**, a transfer transistor is formed over the second substrate, wherein the transfer transistor is electrically coupled from the photodetector to a floating diffusion node. An example of a drawing illustrating this step can be found, for example, in FIG. **13**.

[0061] At **1912**, a second interconnect structure is formed over the transfer transistor and comprising a third conductive path electrically coupled to the floating diffusion node. An example of a drawing illustrating this step can be found, for example, in FIGS. **13-14**.

[0062] At **1914**, the second interconnect structure is bonded to the first interconnect structure via the first bond layer, wherein the bonding electrically couples the first and third conductive paths together via the bond electrode. An example of a drawing illustrating this step can be found, for example, in FIG. **17**.

[0063] Some embodiments relate to an image sensor, including: a photodetector; and a pixel circuit, comprising: a floating diffusion node and an output node; a transfer transistor electrically coupled from the floating diffusion node to the photodetector; a source follower transistor comprising a gate electrode electrically coupled to the floating diffusion node; a row select transistor electrically coupled from a source/drain region of the source follower transistor to the output node; and a capacitor electrically coupled from the output node to the floating diffusion node. In some embodiments, the image sensor further includes: a first integrated circuit (IC) chip accommodating the photodetector, the floating diffusion node, and the transfer transistor; and a second IC chip bonded to the first IC chip at an interface and accommodating the source follower transistor, the row select transistor, and the output node, where the capacitor comprises individual electrodes at the interface. In some embodiments, the interface comprises a metal-to-metal interface and a dielectric-to-dielectric interface. In some embodiments, the image sensor further includes a third IC chip bonded to the second IC chip and separated from the first IC chip by the second IC chip, where the third IC chip comprises an image signal processor electrically coupled to the output node. In some embodiments, the capacitor comprises a first electrode and a second electrode respectively electrically coupled to the output node and the floating diffusion node, and where the first electrode extends in a closed path around the second electrode. In some embodiments, the photodetector and the pixel circuit form a pixel, which repeats in a plurality of rows and a plurality of columns, and where the image sensor further includes: an output line elongated along a first column of the plurality of columns and electrically coupled to the output node of each pixel in the first column. In some embodiments, the source follower transistor has a gain less than 1.

[0064] Other embodiments relate to an image sensor, including: a source follower transistor and a row select transistor on a first substrate, wherein the row select transistor is electrically coupled from the source follower transistor to an output node; a photodetector and a floating diffusion node in a second substrate; a transfer transistor on the second substrate and electrically coupled from the photodetector to the floating diffusion node; a first interconnect structure and a second interconnect structure between the first and second substrates; and a bond structure between the first and second interconnect structures, wherein the bond structure comprises a shield electrode and a bond electrode surrounding the shield electrode; wherein the first interconnect structure electrically couples the shield electrode to the output node, and wherein the second interconnect structure electrically couples the bond electrode to the floating diffusion node. In some embodiments, the first interconnect structure comprises a plurality of wire levels and a plurality of via levels alternatingly stacked from the bond structure towards the first substrate, and wherein the shield electrode has a thickness that is greater than a thickness that a wire level of the plurality of wire levels has. In some embodiments, the bond electrode and the shield electrode form a capacitance negating parasitic capacitance at the output node. In some embodiments, the image sensor further includes: a plurality of pixels in a plurality of rows and a plurality of columns, where the plurality of pixels comprise a first pixel formed in part by the photodetector, the row select transistor, and the source follower transistor, and where the bond structure comprises a plurality of bond electrodes, including the bond electrode, individual to the plurality of pixels. In some embodiments, the shield electrode extends in a closed path around the bond electrode to separate the bond electrode from each other bond electrode of the plurality of bond electrodes. In some embodiments, the first pixel is in a first column of the plurality of columns, and where the shield electrode extends in a closed path around multiple bond electrodes of the plurality of bond electrodes that correspond to pixels in the first column.

[0065] Yet other embodiments relate to a method of forming an image sensor, including: forming an output stage on a first substrate and comprising a row select transistor electrically coupled from a source follower transistor to an output node; forming a first interconnect structure over the output stage, where the first interconnect structure comprises a first conductive path electrically coupled to a gate electrode of the source follower transistor and a second conductive path electrically coupled to the output node; forming a first bond layer over the first interconnect structure, where the first bond layer comprises a bond electrode and a shield electrode surrounding the bond electrode, and where the shield electrode is formed electrically coupled to the second conductive path; forming a photodetector in a second substrate; forming a transfer transistor over the second substrate, where the transfer transistor is electrically coupled from the photodetector to a floating diffusion node; forming a second interconnect structure over the transfer transistor and comprising a third conductive path electrically coupled to the floating diffusion node; and bonding the second interconnect structure to the first interconnect structure via the first bond layer, where the bonding electrically couples the first and third conductive paths together via the bond electrode. In some embodiments, the bond electrode and the shield electrode form a capacitor electrically coupled from the output node to the floating diffusion node. In some embodiments, the method further comprises: forming a second bond layer over the second interconnect structure, where the second bond layer comprises an additional bond electrode and an additional shield electrode, where the additional bond electrode is electrically coupled to the third conductive path, and where the additional bond electrode and the additional shield electrode are bonded respectively to the bond electrode and the shield electrode during the bonding of the second interconnect structure to the first interconnect structure. In some embodiments, the additional bond electrode has a same top geometry as the bond electrode, and wherein the additional shield electrode has a same top geometry as the shield electrode. In some embodiments, the method further comprises forming a conversion gain circuit concurrently with forming the output stage and comprising a first conversion gain transistor, where the first conductive path is electrically coupled to a gate electrode of the first conversion gain transistor. In some embodiments, the method further comprises: forming a third interconnect structure on a backside of the first substrate, where the output stage and the first interconnect structure are formed on a frontside of the first substrate, opposite the backside of the first substrate; and forming a through substrate via (TSV) extending through the first substrate to the third interconnect structure, wherein the first interconnect structure is formed electrically coupled to the TSV. In some embodiments, the method further comprises bonding an integrated circuit (IC) chip to the third interconnect structure on the backside of the first substrate, where the IC chip comprises an image signal processing circuit.

[0066] It will be appreciated that in this written description, as well as in the claims below, the terms “first”, “second”, “second”, “third” etc. are merely generic identifiers used for ease of description to distinguish between different elements of a figure or a series of figures. In and of themselves, these terms do not imply any temporal ordering or structural proximity for these elements, and are not intended to be descriptive of corresponding elements in different illustrated embodiments and/or un-illustrated embodiments. For example, “a first dielectric layer” described

in connection with a first figure may not necessarily correspond to a “first dielectric layer” described in connection with another figure, and may not necessarily correspond to a “first dielectric layer” in an un-illustrated embodiment.
[0067] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A device, comprising: a photodetector; and a pixel circuit, comprising: a floating diffusion node and an output node; a transfer transistor electrically coupled from the floating diffusion node to the photodetector; a source follower transistor comprising a gate electrode electrically coupled to the floating diffusion node; a row select transistor electrically coupled from a source/drain region of the source follower transistor to the output node; and a capacitor electrically coupled from the output node to the floating diffusion node.
2. The device according to claim 1, further comprising: a first integrated circuit (IC) chip accommodating the source follower transistor, the row select transistor, and the output node; and a second IC chip bonded to the first IC chip at an interface and accommodating the photodetector, the floating diffusion node, and the transfer transistor, wherein the capacitor comprises individual electrodes at the interface.
3. The device according to claim 2, wherein the interface comprises a metal-to-metal interface and a dielectric-to-dielectric interface.
4. The device according to claim 2, further comprising: a third IC chip bonded to the first IC chip and separated from the second IC chip by the first IC chip, wherein the third IC chip comprises an image signal processor electrically coupled to the output node.
5. The device according to claim 1, wherein the capacitor comprises a first electrode and a second electrode respectively electrically coupled to the output node and the floating diffusion node, and wherein the first electrode extends in a closed path around the second electrode.
6. The device according to claim 1, wherein the photodetector and the pixel circuit form a pixel, which repeats in a plurality of rows and a plurality of columns, and wherein the device further comprises: an output line elongated along a first column of the plurality of columns and electrically coupled to the output node of each pixel in the first column.
7. The device according to claim 1, wherein the source follower transistor has a gain less than 1.
8. A device, comprising: a source follower transistor and a row select transistor on a first substrate, wherein the row select transistor is electrically coupled from the source follower transistor to an output node; a photodetector and a floating diffusion node in a second substrate; a transfer transistor on the second substrate and electrically coupled from the photodetector to the floating diffusion node; a first interconnect structure and a second interconnect structure between the first and second substrates; and a bond structure between the first and second interconnect structures, wherein the bond structure comprises a bond electrode and a shield electrode surrounding the bond electrode; wherein the first interconnect structure electrically couples the shield electrode to the output node, and wherein the second interconnect structure electrically couples the bond electrode to the floating diffusion node.
9. The device according to claim 8, wherein the first interconnect structure comprises a plurality of wire levels and a plurality of via levels alternatingly stacked from the bond structure towards the

first substrate, and wherein the shield electrode has a thickness that is greater than a thickness of a wire level of the plurality of wire levels.

10. The device according to claim 8, wherein the bond electrode and the shield electrode form a capacitance negating parasitic capacitance at the output node.

11. The device according to claim 8, further comprising: a plurality of pixels in a plurality of rows and a plurality of columns, wherein the plurality of pixels comprise a first pixel formed in part by the photodetector, the row select transistor, and the source follower transistor, and wherein the bond structure comprises a plurality of bond electrodes, including the bond electrode, individual to the plurality of pixels.

12. The device according to claim 11, wherein the shield electrode extends in a closed path around the bond electrode to separate the bond electrode from each other bond electrode of the plurality of bond electrodes.

13. The device according to claim 11, wherein the first pixel is in a first column of the plurality of columns, and wherein the shield electrode extends in a closed path around multiple bond electrodes of the plurality of bond electrodes that correspond to pixels in the first column.

14. A method of forming a device, comprising: forming an output stage on a first substrate and comprising a row select transistor electrically coupled from a source follower transistor to an output node; forming a first interconnect structure over the output stage, wherein the first interconnect structure comprises a first conductive path electrically coupled to a gate electrode of the source follower transistor and a second conductive path electrically coupled to the output node; forming a first bond layer over the first interconnect structure, wherein the first bond layer comprises a bond electrode and a shield electrode surrounding the bond electrode, and wherein the shield electrode is formed electrically coupled to the second conductive path; forming a photodetector in a second substrate; forming a transfer transistor over the second substrate, wherein the transfer transistor is electrically coupled from the photodetector to a floating diffusion node; forming a second interconnect structure over the transfer transistor and comprising a third conductive path electrically coupled to the floating diffusion node; and bonding the second interconnect structure to the first interconnect structure via the first bond layer, wherein the bonding electrically couples the first and third conductive paths together via the bond electrode.

15. The method according to claim 14, wherein the bond electrode and the shield electrode form a capacitor electrically coupled from the output node to the floating diffusion node.

16. The method according to claim 14, further comprising: forming a second bond layer over the second interconnect structure, wherein the second bond layer comprises an additional bond electrode and an additional shield electrode, wherein the additional bond electrode is electrically coupled to the third conductive path, and wherein the additional bond electrode and the additional shield electrode are bonded respectively to the bond electrode and the shield electrode during the bonding of the second interconnect structure to the first interconnect structure.

17. The method according to claim 16, wherein the additional bond electrode has a same top geometry as the bond electrode, and wherein the additional shield electrode has a same top geometry as the shield electrode.

18. The method according to claim 14, further comprising: forming a conversion gain circuit concurrently with forming the output stage and comprising a first conversion gain transistor, wherein the first conductive path is electrically coupled to a gate electrode of the first conversion gain transistor.

19. The method according to claim 14, further comprising: forming a third interconnect structure on a backside of the first substrate, wherein the output stage and the first interconnect structure are formed on a frontside of the first substrate, opposite the backside of the first substrate; and forming a through substrate via (TSV) extending through the first substrate to the third interconnect structure, wherein the first interconnect structure is formed electrically coupled to the TSV.

20. The method according to claim 19, further comprising: bonding an integrated circuit (IC) chip to the third interconnect structure on the backside of the first substrate, wherein the IC chip comprises an image signal processing circuit.